# Serial Peripheral Interface (SPI) in STM32F446RE

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# WHAT IS SPI?

## SPI(Serial Peripheral Interface):

SPI is a serial communication protocol used to exchange data between a microcontroller and external devices. It operates using a Master–Slave architecture, where the Master controls the communication by generating the clock signal, and the Slave responds to the Master's commands.

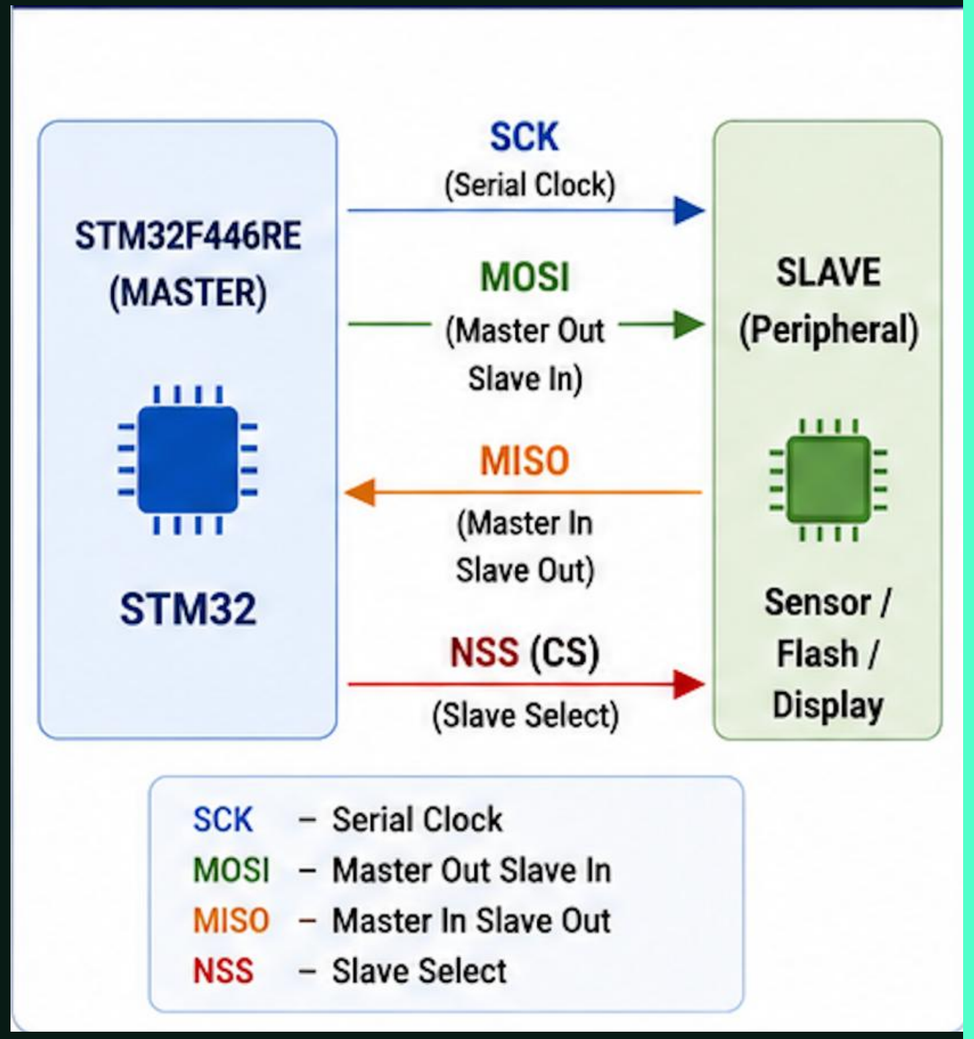
SPI supports full-duplex communication, meaning that both devices can transmit and receive data simultaneously. This makes SPI a fast, simple, and efficient communication protocol, especially for high-speed embedded applications.

# Features

- 4 SPI peripherals (SPI1, SPI2, SPI3, SPI4)
- Supports Master and Slave modes
- Full-duplex, Half-duplex, and Simplex communication
- Synchronous communication (provide clock with data)
- 8-bit or 16-bit data frame format
- Programmable baud rate (clock speed)
- Hardware CRC for error detection
- DMA support for high-speed data transfer
- Interrupt support for efficient communication

# Communication Lines:

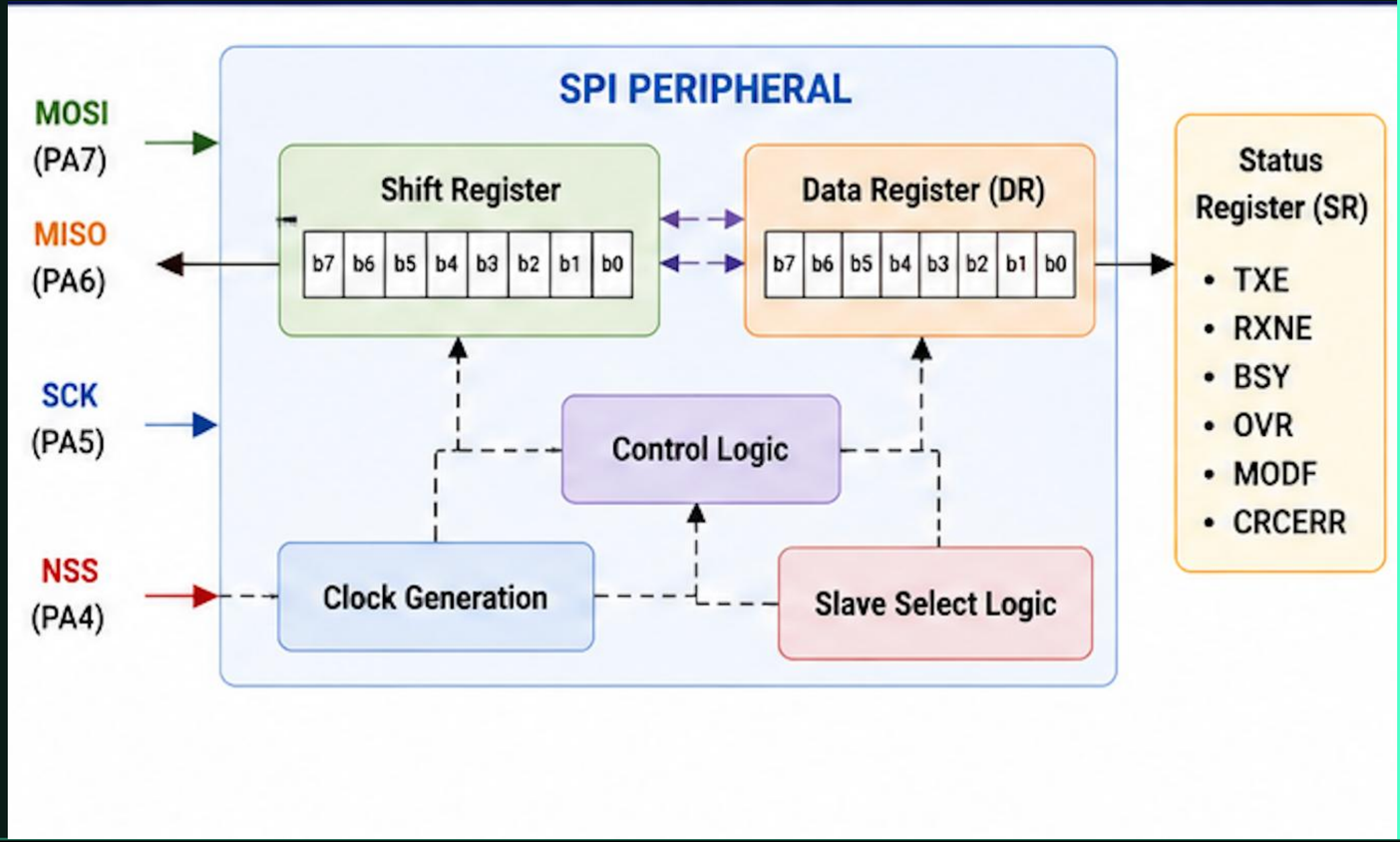
| Signals | Function |
|---|---|
| SCK | Serial Clock generated by Master |
| MOSI | Master sends data to Slave |
| MISO | Slave sends data to Master |
| NSS(CS) | Selects the Slave device |



# SPI REGISTERS

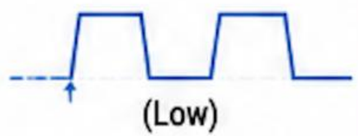
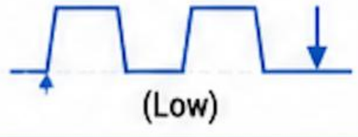
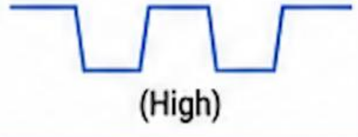
## Key Registers

1. **SPI_CR1** is the main configuration register. It sets the operating mode, baud rate, clock polarity, clock phase, and enables the SPI.
2. **SPI_CR2** contains additional configuration settings, such as interrupt and DMA options.
3. **SPI_SR** is the status register. It indicates whether the transmitter or receiver is ready and whether SPI is busy.
4. **SPI_DR** is the data register used to send and receive data.



# SPI Clock Modes

- CPOL – Clock Polarity
- CPHA – Clock Phase
- Four modes: Mode 0, 1, 2, 3
- Master and Slave must use the same mode

| Mode | CPOL | CPHA | Clock Idle State | Sampling Edge |
|---|---|---|---|---|
| 0 | 0 | 0 |  (Low) | First (Leading) Edge |
| 1 | 0 | 1 |  (Low) | Second (Trailing) Edge |
| 2 | 1 | 0 |  (High) | First (Leading) Edge |
| 3 | 1 | 1 |  (High) | Second (Trailing) Edge |

* Master and Slave must use the same CPOL, CPHA (Mode).

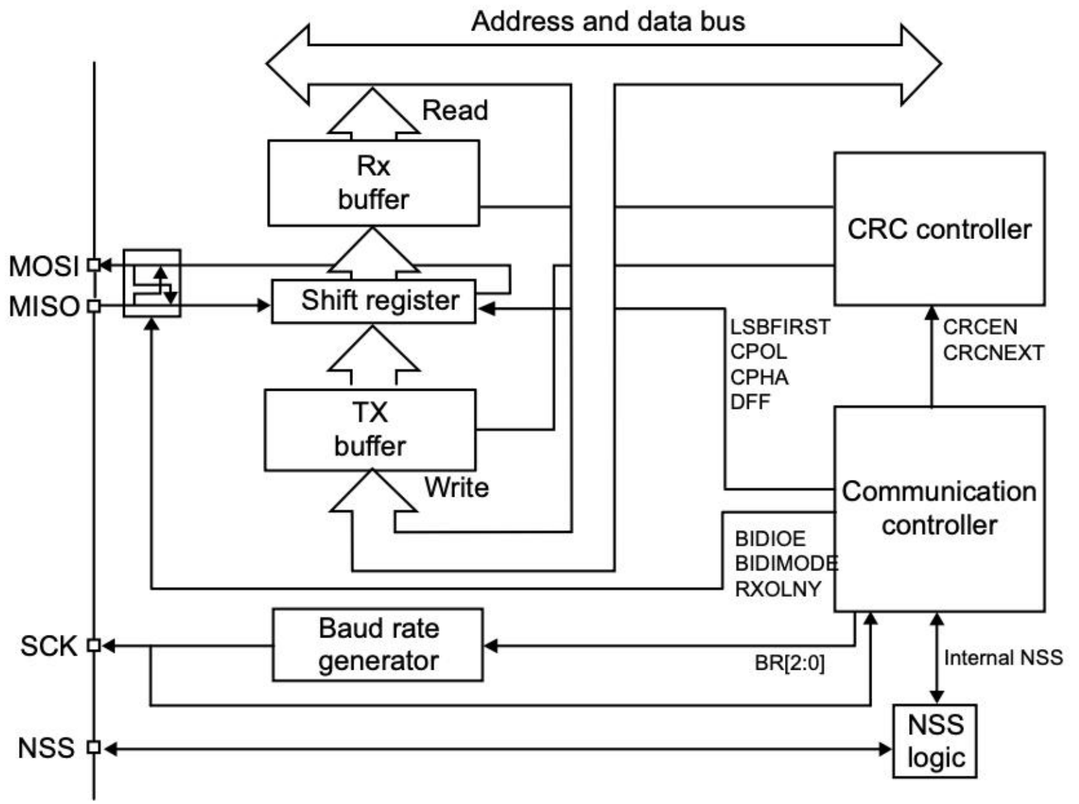
# HOW SPI WORKS

## Steps in Communication Process

1. Master pulls **NSS LOW**.
2. Master generates **clock (SCK).**
3. Data is sent through **MOSI.**
4. Slave sends data through **MISO.**
5. After transmission, Master sets **NSS HIGH.**

**Key Point:** SPI transmits and receives data at the same time **(Full-Duplex)**

**Figure 303. SPI block diagram**

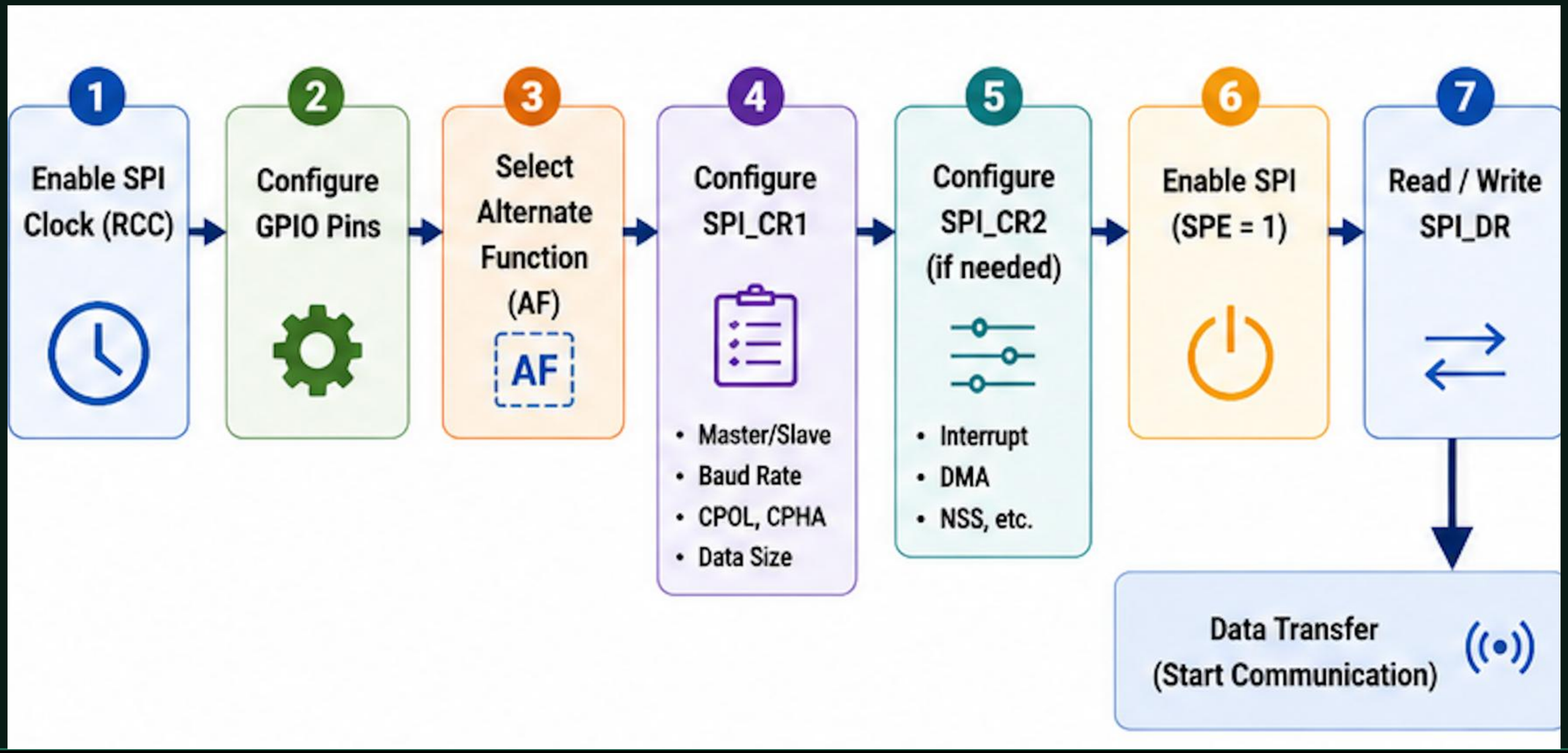


# SPI Initialization–(Bare Metal)

The basic steps are:

- Enable the SPI peripheral clock.
- Configure the GPIO pins for Alternate Function mode.
- Configure SPI settings such as Master mode, baud rate, and clock mode.
- Enable SPI by setting the SPE bit.
- Write data to the SPI_DR register to begin communication.

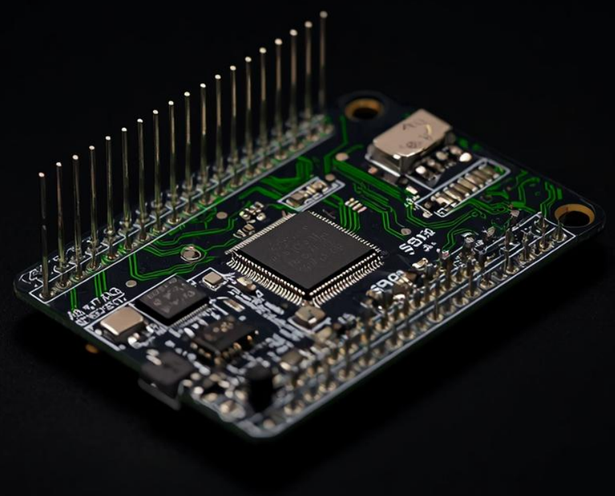
# SPI Initialization Flowchart



# PIN CONFIGURATION

Default SPI1 Pin Mapping

| Signal | Pin |
|---|---|
| SCK | PA5 |
| MOSI | PA6 |
| MISO | PA7 |
| NSS(CS) | PA4 |



## Applications

SPI Applications

- Sensors — temperature, pressure, motion, etc.
- LCD / TFT Displays — sending display data
- SD Cards — reading and writing data
- Flash Memory — storing and retrieving data
- ADC / DAC — digital and analog data conversion
- Wireless Modules — communication with external modules
- EEPROM / Memory Devices — non-volatile data storage

Key Point: SPI is mainly used for fast and efficient communication between the STM32F446RE and external peripheral devices.

## Advantages:

- High-speed data communication
- Full-duplex transfer (simultaneous send & receive)
- Simple hardware interface with few lines
- DMA support for efficient data handling

## Disadvantages:

- Requires more wires than I²C
- No built-in device addressing (each slave needs its own NSS/CS line)
- Less suitable when many slave devices are connected

# THANK YOU